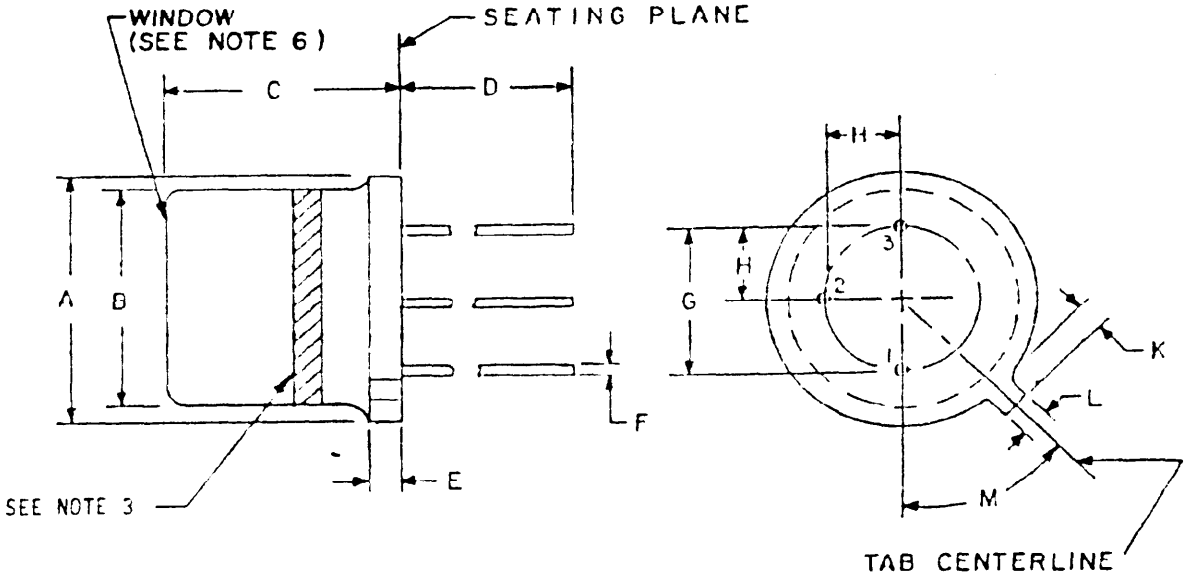
MILITARY SPECIFICATION SHEET

SOURCES, LED, FIBER OPTIC, TYPE A, CLASS 1, STYLE 3

This specification is approved for use by Departments and Agencies of the Department of Defense.

The requirements for acquiring the fiber optic source described herein shall consist of this specification and the latest issue of MIL-S-24622.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. A yellow band, a yellow dot or a yellow mark shall be provided on the LED case to specify that this is a fiber optic device.
4. Pin 1, which is closest to the tab, shall be the anode conductor. Pin 2 is the cathode and pin 3 is internally connected to the case.
5. The lead conductor's positional tolerance to the seating plane shall be

⌖	⌀0.36 (0.014)	Ⓜ	T	A Ⓜ	L Ⓜ

where A and L are datums and T is a datum plane.

6. The optical window shall not extend beyond the face of the can by more than 1 mil. There shall be no bubbles greater than 5 mil within the 40 mil diameter window.

FIGURE 1. LED source.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

Letter symbol	Inches		Millimeters	
	Min	Max	Min	Max
A	0.209	0.230	5.31	5.84
B	0.183	0.185	4.65	4.70
C	0.123	0.129	3.12	3.28
D	0.500		12.70	
E		0.030		0.76
F	0.016	0.019	0.41	0.48
G	0.100 T. P.		2.54 T. P.	
H	0.050 T. P.		1.27 T. P.	
J	0.050 T. P.		1.27 T. P.	
K	0.033	0.048	0.84	1.22
L	0.039	0.046	0.99	1.17
M	45° T. P.		45° T. P.	

FIGURE 1. LED source - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Lead conductors: Lead conductors shall be Alloy-42 or Kovar (American wire gauge 24) with a hot solder dip finish.

Mounting: Fiber optic alignment sleeves will be provided with MIL-C-83522/4 fiber optic connector receptacles for active devices.

Operating temperature: -65°C to +125°C.

Storage temperature: -65°C to + 150°C.

Terminal strength: Test condition E. The leads shall be bent 3 times with an 8 oz. lead bend for 15 seconds.

Electrical and optical requirements: See table I.

Identification marking:

Part number: See table II.

Color band: A yellow band, yellow dot or a yellow mark shall be included on the assembly case as shown on figure 1.

Marking: Marking size for source type-designation shall be 1 mm in height, and the manufacturer's designating symbol shall be identifiable without need for visual magnification.

Screening: Screening shall be conducted on 100 percent of all devices prior to groups A, B, and C inspection and shall be in accordance with the methods specified in table II.

Group A inspection: Group A inpsection shall consist of the examinations and tests specified in table III.

Group B inspection: Group B inpsection shall consist of the examinations and tests specified in table IV.

Group C inspection: Group C inpsection shall consist of the examinations and tests specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE I. Source characteristics and part number. 1/

Source part number M24622 /02-	Forward current I_F(MA)	Forward voltage V_F(V) 2/	Total power P_T (mW) 3/	Reverse voltage $V_{(BR)R}$(V) 4/	Reverse current I_R (mA)	Thermal resistance R_G (°C/W) 5/	Peak wavelength λ_p (nm) 7/	Wavelength drift λ_d (nm/°C)	Spectral width $\Delta\lambda$ (nm) 8/	Power amplitude (µW) 9/	Frequency response MHz 13/	Pulse times (ns) 14/ t_r	Pulse times (ns) 14/ t_f
-01	100 mA max.	1.85 ±.35	250	2 min	1.0 max.	440 or 225 6/	850 ±20	.245	50	11-22 10/	100	4	6
										75-150 10/			
-02										27-56 11/	20	20	22
										180-360 11/			
-03										21-42 12/	100	4	6
										135-270 12/			

See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

1/ Unless otherwise specified, the ambient temperature shall be 30°C + 5°C. Also, the sources covered by this specification sheet shall meet the dimensional requirements as specified herein. However, for reference only, these sources are similar to a TO-52 packaged device.

2/ The forward voltage shall be determined at a forward current of 100 mA.

3/ The total device power dissipation at an ambient temperature (T_A) of 25°C. Above 25°C the power shall be derated at 2.5 mW/°C.

4/ Reverse voltage $V_{(BR)R}$ shall be 2.0 V min. when the reverse current is set at 100 µA.

5/ Thermal resistance (R_G) is the junction to ambient value when installed in the alignment bushing and mounted in applicable MIL-C-83522 connector.

6/ For a device in free air the R_G shall be 440°C/W. For a device mounted in a MIL-C-83522 connector, the R_G shall be 225°C/W.

7/ λ_p is the wavelength at peak output amplitude.

8/ $\Delta\lambda$ is the spectral width between half amplitude points.

9/ The power amplitude shall be measured at the end of 1 meter of the specified cable when the source is installed in the alignment bushing and connected to the cable with the applicable MIL-C-83522 connector. The power amplitude is found with a forward current of 100 mA and a wavelength of 850 nm.

10/ The power amplitude of M24622/02-01 when connected with a 50/125 fiber constructed in accordance with DOD-F-49291 shall be 11-22 µW and 75-150 µW when connected with a 100/140 fiber constructed in accordance with DOD-F-49291.

11/ The power amplitude of M24622/02-02 when connected with a 50/125 fiber constructed in accordance with DOD-F-49291 shall be 27-42 µW and 180-360 µW when connected with a 100/140 fiber constructed in accordance with DOD-F-49291.

12/ The power amplitude of M24622/02-03 when connected with a 50/125 fiber constructed in accordance with DOD-F-49291 shall be 21-42 µW and 135-270 µW when connected with a 100/140 fiber constructed in accordance with DOD-F-49291.

13/ Frequency response value corresponds to the -3dB power output level with a forward current of 80 mA and over a frequency range of 10 to 110 MHz.

14/ t_r/t_f values are taken at the 10% and 90% amplitude points with a forward current of 100 mA dc. The rise and fall times for M24622/02-01 as well as -03 are 4 and 6 ns, respectively. The rise and fall times for M24622/02-02 are 20 and 22 ns, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE II. Screening inspections.

Screen inspection (100%)	MIL-STD-750 method	Details
Internal visual (precap) inspection	2073	Ambient
High temperature (nonoperating) life (LTPD)	1032	24 hours minimum at T = +125°C for type A devices and +85°C for type B devices.
Thermal shock (temperature cycling)	1051	Test condition A-1, low temperature = -46°C, high temperature = +125°C for type A and +85°C for type B.
Constant acceleration	2006	Y_1 direction at 20,000 G at T_C = 25°C. The one minute hold time requirement shall not apply.
Hermetic seal a. Fine leak rate 1/ b. Gross leak rate	1071	 Type A only, condition G or H. Type A and type B, condition C or D. Fluid temperature = +85°C.
Forward voltage (V_F) Power launched amplitude (P_L)	4011 MIL-STD-1678 method 6010	I_F = 100 mA Procedure I
Steady-state power burn-in	1038	Maximum rated I_F for 96 hours T_J = 100°C min for type A and +70°C min for type B. Condition B.
ΔV_F 2/ ΔP_L		A delta of 25% for V_F and 20% for P shall be permitted from initial values after burn-in
External visual and mechanical inspection 3/	2071	Ambient

1/ Sealing level type B sources may omit this test.
2/ If the quantity in the lot which fails the delta limits exceeds 10 percent than the lot fails. If the percent defective exceeds 20 percent, the lot shall not be acceptable for any level. For those lots resubmitted for burn-in the PDA shall not exceed 3 percent.
3/ Visual and mechanical inspection shall be performed on all devices after marking has been completed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection.

Industry sublots	MIL-STD-750 method	Conditions	LTPD
Subgroup 1			
External visual and mechanical inspection	2071		5
Subgroup 2		To be performed at 25°C. I_F = 100 mA	
Forward voltage	4011		5
Power launched amplitude	MIL-STD-1678 method 6010	Procedure I	
Subgroup 3			
Pulse rise and fall time -or- Frequency response		To be performed at +25°C	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE IV. Group B inspection. 1/

Inspection subgroups	MIL-STD-750 method	Conditions	LTPD
Subgroup 1			
Solderability 2/	2026		15
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		10
a. Fine leak		Type A, test condition G or H	
b. Gross leak		Type A and type B, test condition C or D, fluid temp = +85°C	
Electrical and optical measurements			
See group A, subgroup 2			
Subgroup 3 3/			
Steady state operation life (LTPD)	1027	340 hours, bias conditions as specified	5
Electrical and optical measurements			
See group A, subgroup 2			
Subgroup 4			
Decap internal visual design verification	2075		1 device/0 failures for each lot
Bond strength	2037	The sample shall include a min of 3 devices	10(c = 0)
Subgroup 5			
High temperature (non-(operating) life (LTPD)	1032		7
Electrical and optical measurements			
See group A, subgroup 2			

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ The LTPD for solderability test applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

3/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340 hour life tests may be continued on test to 1,000 hours in order to satisfy the group C life test requirements. In such cases, either the 340 hour end point measurements must be made as a basis for group B lot acceptance or the 1,000 hours end point measurements shall be used as the basis for both group B and group C acceptance.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE V. Group C inspection.

Inspection subgroups	MIL-STD-750 method	Conditions 1/	LTPD
Subgroup 1			
Physical dimensions 2/	2066	See figure 1	15
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition A	10
Terminal strength	2036	Test condition E	
Hermetic seal	1071		
a. Fine leak		Type A, test condition G or H	
b. Gross leak		Type A and B, Test Condition C or D, Fluid temperature = +85°C.	
Moisture resistance	1021		
Electrical and optical measurements		As specified	
See group A, subgroup 2			
Subgroup 3			
Shock	2016	Non-operating, 1500 G's, 0.5 ms, 5 blows in each orientation: X1, Y1, and Z1	
Vibration	2056		
Constant acceleration	2006	One minute min in each orientation, X1, Y1, and Z1 at 20,000 G min T_C = 25°C	10
Electrical and optical measurements			
See group A, subgroup 2			
Subgroup 4 2/			
Salt atmosphere (corrosion)	1041		15
Subgroup 5			
Thermal resistance	4081		15
Peak optical wavelength			
Subgroup 6 3/			
Steady-state operation life (λ)	1025		λ = 10
Electrical and optical measurements			
See group A, subgroup 2			

1/ See requirements and test method paragraphs for complete exposure conditions.
2/ Electrical and optical reject devices from the same inspection lot may be used for all subgroups when electrical and optical end point measurements are not required.
3/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 340 hour life tests may be continued on test to 1,000 hours in order to satisfy the group C life test requirements. In such cases, either the 340 hour end-point measurements must be made as a basis for group B lot acceptance or the 1,000 hours end-point measurements shall be used for both groups B and C acceptance.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:24Z
Check the source to verify that this is the currentversion before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities
Navy - AS, EC
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES
NASA - NA

User activities:
Navy - CG, MC
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6030-0017)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.